MACHINE LEARNING–BASED EYE STRAIN AND DIGITAL FATIGUE PREDICTION SYSTEM FOR COLLEGE STUDENTS USING SCREEN TIME DATA

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CHAPTER 1 - INTRODUCTION

The problem and its background refer to the context and description of an issue or challenge that requires attention or investigation. It sets the stage for understanding the problem's significance, its underlying causes, and the reasons why it needs to be addressed. Providing a clear background helps to establish the context and enables researchers or problem solvers to develop appropriate strategies and solutions.

Background of the Study

The use of digital devices has become an essential part of daily life, especially among students. Laptops, smartphones, tablets, and other electronic devices are widely used for academic, social, and recreational activities. As technology continues to advance, students are increasingly exposed to prolonged screen time, which has led to a growing health concern known as digital eye strain, also referred to as computer vision syndrome.

Digital eye strain is characterized by various symptoms such as dry eyes, blurred vision, headaches, and neck pain. In addition, extended screen exposure may lead to digital fatigue, which includes mental exhaustion, reduced concentration, and decreased productivity. College students are particularly vulnerable to these conditions due to their intensive use of digital devices for online classes, research, assignments, and communication.

As the reliance on digital technology continues to grow, there is a need to develop solutions that can help monitor and manage screen usage. The emergence of machine learning has provided new opportunities in health monitoring and predictive analysis. Machine learning algorithms can analyze patterns in data and identify potential risks before

they become severe.

For this reason, the researchers aim to develop a system called “Web-Based Screen Time Analyzer for Predicting Eye Strain and Digital Fatigue.” The system will collect screen time data and analyze it using machine learning algorithms to assess the risk level of users. Through this system, users can receive early warnings and recommendations to help reduce the negative effects of prolonged screen exposure.

Project Context

In the context of the researchers’ institution, the use of digital devices has become an essential part of students’ academic activities. College students regularly utilize laptops, smartphones, and tablets for attending online or hybrid classes, completing assignments, conducting research, and communicating with instructors and classmates. As a result, prolonged screen exposure has become a daily routine for many students within the school.

Students in the institution often spend extended hours in front of digital screens not only for academic purposes but also for social media and entertainment. This continuous exposure increases the likelihood of experiencing eye strain and digital fatigue. However, despite the presence of these issues among students, there is currently no existing system within the school that can assess and predict the risk of eye strain and digital fatigue based on screen usage behavior.

Existing digital wellness tools are available, but they are not commonly utilized by students due to the need for installation, limited features, or lack of awareness. Moreover, these tools typically provide only basic monitoring and do not offer predictive analysis or personalized recommendations based on user data.

Furthermore, there is a lack of systems that utilize machine learning to analyze screen time data and predict the risk of eye strain and digital fatigue among college students. Most available solutions are general-purpose applications and do not specifically address the needs and usage patterns of students.

The proposed system entitled “Machine Learning–Based Eye Strain and Digital Fatigue Prediction System for College Students Using Screen Time Data” aims to address these gaps by providing an accessible web-based platform that does not require installation. While the study will be conducted within the researchers’ institution, the system is designed to be applicable and usable by college students in general. By collecting and analyzing screen time data, the system will predict risk levels and provide early warnings and recommendations to promote healthier screen usage habits among students.

Purpose and Description

The study was conducted in order to help college students manage their screen time and reduce the risk of eye strain and digital fatigue by developing a Machine Learning–Based Eye Strain and Digital Fatigue Prediction System. The system has the following capabilities: [1] it can collect and store daily screen time data from users; [2] it can gather self-reported symptoms related to eye strain and digital fatigue; [3] it can analyze the collected data using machine learning algorithms; [4] it can provide risk level predictions and basic recommendations through a user-friendly web interface.

Once the system is implemented, it is expected to benefit the following individuals:

College Students. They will benefit from the system by gaining awareness of their screen usage habits and understanding the potential risks associated with prolonged

screen time. The system can help them adjust their behavior to reduce discomfort and improve productivity.

Researchers. The development of this study provides knowledge and experience in applying machine learning and web technologies. It enhances technical skills and contributes to understanding how technology can be used in health-related applications.

Future Researchers. This study can serve as a reference for future research related to digital health monitoring and machine learning applications. Future researchers can improve the system by addressing its limitations and integrating more advanced features.

Statement of the Problem

College students represent one of the most screen-dependent populations in today's digital era. Their academic responsibilities require continuous use of laptops, smartphones, and tablets for online learning, research, and assignments, while social media and entertainment further increase their daily screen exposure. As a result, prolonged and often uninterrupted screen usage has contributed to the increasing occurrence of eye strain and digital fatigue among students. Despite this growing concern, there is still a lack of accessible and intelligent systems that can effectively predict and manage these conditions using screen time data.

1. The planning phase of the proposed web-based system, titled "Machine Learning–Based Eye Strain and Digital Fatigue Prediction System for College Students Using Screen Time Data," requires a comprehensive analysis of the specific needs of college students in monitoring digital wellness. The study aims to design a website that includes proper requirement analysis, system architecture, database design, and suitable machine learning techniques to accurately predict eye strain and

digital fatigue. Without a well-planned framework, the website may fail to provide reliable, user-centered, and preventive digital wellness support for students.

2. The development phase of the proposed website focuses on creating an integrated and user-friendly platform that collects and processes screen time data to generate predictive results. The system must effectively combine data collection, preprocessing, machine learning model implementation, and a responsive web-based interface. Existing approaches often only provide basic monitoring features such as screen time tracking and usage reports. Therefore, this study aims to develop a website that not only monitors usage but also predicts the risk level of eye strain and digital fatigue while providing preventive recommendations for college students.

3. The evaluation phase of the proposed website seeks to determine the system's functionality, usability, prediction accuracy, and effectiveness in helping students manage eye strain and digital fatigue. The study will assess whether the machine learning model can generate reliable predictions based on screen time data and whether the website interface is accessible and convenient for users. Proper evaluation is necessary to ensure that the developed system meets the intended objectives and provides meaningful support for students' digital well-being.

4. The study addresses the limited implementation of machine learning in web-based digital wellness systems designed specifically for college students. Most existing systems are reactive, meaning they only provide feedback after symptoms of eye strain and fatigue occur. The proposed website aims to provide a proactive solution by predicting possible eye strain and digital fatigue risks before symptoms worsen. Through early prediction and preventive recommendations, the system can help students improve screen usage habits and maintain better digital health.

Objectives of the Study

This study aims to develop a Machine Learning–Based Eye Strain and Digital Fatigue Prediction System for College Students Using Screen Time Data. It focuses on addressing the planning, development, and evaluation of the system to ensure accurate prediction and effective usability. Specifically, this study seeks to achieve the following objectives:

1. To plan the system by identifying user requirements, designing the system architecture, and selecting appropriate machine learning algorithms suitable for predicting eye strain and digital fatigue.

2. To develop the system by collecting and preprocessing screen time and symptom data, implementing machine learning models for prediction, developing a web-based user interface, integrating an AI chatbot and admin panel, and combining all system components into a functional application.

3. To evaluate the system by testing the accuracy and performance of the machine learning model, assessing the usability of the web-based system, and determining its effectiveness in predicting eye strain and digital fatigue.

4. To determine the applicability of machine learning–based prediction by analyzing how screen time data can be used to identify risk levels of eye strain and digital fatigue among college students and evaluating its usefulness as a preventive digital wellness tool.

Scope and Limitation

This study focuses on the development of a web-based Machine Learning-Based Eye Strain and Digital Fatigue Prediction System intended for college students who are frequently exposed to digital screens for academic and personal activities. The system utilizes machine learning algorithms, specifically Random Forest and XGBoost, to analyze

screen time behavior and self-reported symptoms in order to predict the risk level of eye strain and digital fatigue.

The study covers the planning, development, implementation, and evaluation of the proposed system. The system collects user-inputted screen time data and symptoms related to eye strain and digital fatigue, processes the collected information, and generates predictive results and recommendations. The system also includes an AI chatbot feature that provides general information and guidance regarding healthy screen usage habits. In addition, an admin panel is integrated to manage user records, monitor prediction results, and oversee system operations. The system utilizes Supabase for database management and data storage.

Furthermore, the study includes the development of a user-friendly web interface accessible through standard internet browsers. The proposed system is evaluated based on functionality, usability, and machine learning model performance.

However, the study is limited to self-reported data and manually entered screen time information, which may affect the accuracy of the collected dataset. The system does not include advanced medical assessment tools such as eye-tracking devices, biometric sensors, or clinical examinations. Additionally, the AI chat-bot is intended only for informational and assistive purposes and does not provide medical diagnosis or professional healthcare advice. The study is also limited to the development of the system and the short evaluation period and does not include long-term monitoring of user behavior and eye health conditions.

CHAPTER 2 - REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature and studies that have relevance to the present study. Printed books as well as eBooks, theses, journals, and articles that are either published or unpublished were used. Electronic sources were also used to enhance the related theories and models.

Physiological Indicators and Eye-Tracking Analysis

The nonlinear characteristics of eye movements in detecting visual fatigue have been examined. The study found that eye movement behaviors, such as fixation patterns and saccadic movements, change significantly when visual fatigue occurs. Through nonlinear analysis, these ocular signals were identified as reliable indicators for detecting fatigue levels. This approach provides a scientific basis for using eye-tracking data in fatigue prediction systems. The findings are also relevant to the development of machine learning models, as eye movement features can be used as input data for accurate classification of digital eye strain (Chu et al. 2026).

Redondo et al. (2026) explored tear molecule concentrations as potential biomarkers of digital eye strain. The study identified specific biochemical changes in tear composition that correlate with visual fatigue levels. This physiological approach provides objective evidence for detecting eye strain beyond subjective reporting. The research contributes to the understanding of biological indicators associated with prolonged screen exposure. However, the requirement for laboratory analysis limits real-world application, reinforcing the need for more accessible, non-invasive prediction methods such as machine learning models based on behavioral data.

Li et al. (2021) explored the use of vertical electrooculogram signals as a method for evaluating eye strain during prolonged visual tasks. The study analyzed electrical signals generated by eye movements to determine whether specific patterns could indicate the onset of visual fatigue. Findings showed that measurable changes in ocular muscle activity occur as eye strain develops, suggesting that physiological monitoring can provide objective evidence of fatigue. The research demonstrates that electrooculogram data can serve as a reliable indicator for detecting eye strain in individuals performing continuous digital tasks. Although such physiological measurement methods may require specialized equipment, they offer valuable insights into the biological processes associated with visual fatigue. These findings highlight the potential of integrating physiological indicators with computational techniques, such as machine learning, to improve the accuracy of fatigue detection systems.

Wang et al. (2022) developed a machine learning model that detects eye fatigue using electrooculography (EOG) signals. The study demonstrated that physiological signals from eye movements can be effectively used to classify fatigue levels with high accuracy. However, the use of specialized sensors makes the approach more invasive and less practical for everyday use. This highlights a gap in accessibility, supporting the need for non-invasive, data-driven approaches such as screen time-based prediction systems.

The use of eye-tracking signals to detect visual fatigue has been investigated using machine learning techniques. The study found that indicators such as fixation duration and blink patterns are effective in identifying fatigue. Although the results showed high accuracy, the dependence on eye-tracking devices limits its wider application. This highlights the need to develop non-invasive alternatives that use more accessible data, such as screen usage patterns, to detect visual fatigue (Sun et al. 2024).

Zhao et al. (2025) developed a real-time dual-eye blink detection system using machine learning techniques. The study demonstrated that blink patterns can serve as reliable indicators of visual fatigue, highlighting the importance of eye activity monitoring in fatigue detection systems.

Nagarajan et al. (2024) analyzed blink rate patterns using eye-tracking technology and found that reduced blinking is strongly associated with increased eye strain. The findings provide physiological evidence supporting the relationship between blink behavior and visual fatigue.

Lee et al. (2021) developed a classification system that utilizes eye blink rate and postural behavior data to detect fatigue levels. The study demonstrated that combining behavioral and physiological indicators improves the accuracy of fatigue detection, supporting the development of integrated predictive systems.

Machine Learning and AI Applications

A multimodal artificial intelligence framework has been proposed for the non-invasive detection of digital eye strain using thermal imaging and behavioral data. The system combined facial temperature patterns with user behavior to identify levels of visual fatigue. The results showed that using multiple data sources improved the accuracy of fatigue detection. This study demonstrates the effectiveness of AI in analyzing both physiological and behavioral information. However, the use of thermal imaging devices may limit accessibility, highlighting the need for simpler models that rely on screen time data for more practical implementation (Persiya & Sasithradevi, 2025).

Dai et al. (2026) presented an expert consensus on classification and annotation methods for dry eye imaging in artificial intelligence applications. The study established

standardized procedures for data labeling, quality control, and model validation in ocular research. This work ensures consistency and reliability in AI-based eye health systems. The guidelines support the development of accurate machine learning models by emphasizing proper dataset preparation. This contributes to the foundation of building reliable prediction systems for digital eye strain.

A classification model has been proposed to predict visual fatigue using physiological responses under different lighting conditions. The study showed that combining environmental factors with physiological data can improve prediction accuracy. However, the use of sensor-based data collection limits its application in real-world settings. This finding supports the need to develop alternative machine learning approaches that use more accessible data, such as screen time, making the system more practical and scalable for student use (Jung et al. 2024).

Aggarwal et al. (2022) developed a machine learning-based approach to detect eye strain caused by the use of electronic devices. The study utilized behavioral and symptom-related data to classify levels of eye strain, demonstrating that machine learning models can effectively identify patterns associated with visual fatigue. This supports the potential of data-driven approaches in predicting eye strain using accessible user-generated data.

Saxena et al. (2022) investigated the use of machine learning libraries to identify digital eye strain symptoms. The study demonstrated that classification algorithms can accurately detect symptoms based on structured datasets derived from user responses. The findings highlight the feasibility of using survey-based data in developing predictive models for eye strain.

Jia et al. (2024) proposed a multimodal visual fatigue assessment model that com-

bines backpropagation neural networks and XGBoost algorithms. The study achieved high prediction accuracy by integrating multiple data sources, including physiological and behavioral indicators. This demonstrates the effectiveness of hybrid machine learning models in improving the accuracy of visual fatigue prediction systems.

Dhankar et al. (2025) introduced Lumina AI, a real-time eye wellness monitoring system that utilizes computer vision and machine learning techniques. The system continuously analyzes eye activity to detect signs of fatigue, demonstrating the potential of AI-driven solutions in promoting eye health and preventing digital eye strain.

Annapurna et al. (2023) proposed a machine learning-based approach for predicting smartphone vision syndrome using behavioral data such as screen time and usage patterns. The study demonstrated that such data can effectively predict visual fatigue, supporting the use of behavioral indicators in machine learning models.

Shi et al. (2024) proposed an attentional BiLSTM-based model for predicting smartphone-related visual fatigue. The study demonstrated that deep learning techniques can effectively predict fatigue levels based on user behavior data. This supports the integration of advanced machine learning approaches in predictive systems.

Screen Time Behavior and Mitigation

Redondo et al. (2025) investigated the impact of break schedules on digital eye strain symptoms and ocular accommodation during prolonged near work. The study found that structured and regular breaks significantly reduce symptoms such as blurred vision, dryness, and discomfort. Improved ocular accommodation was also observed when proper rest intervals were implemented. This research highlights the importance of integrating time-based behavioral data, such as screen duration and break frequency, into predictive

systems. The results support the idea that screen time patterns are critical variables in assessing and preventing digital fatigue among students.

Beeson et al. (2024) analyzed the relationship between prolonged digital tasks and eye strain symptoms. The study found that extended screen exposure leads to increased discomfort, reduced visual performance, and decreased productivity. The results indicate a direct connection between screen time duration and fatigue severity. This supports the use of screen time as a key variable in predictive models. The findings are particularly relevant for student populations, where prolonged device usage is common.

Digital eye strain and dry eye symptoms can be mitigated through the implementation of the widely recommended 20-20-20 rule, which advises individuals to take a 20-second break to focus on an object approximately 20 feet away after every 20 minutes of screen use. The application of this strategy has been shown to improve visual comfort and enhance binocular vision performance among individuals who consistently follow the recommended breaks compared to those who do not practice regular rest intervals. Furthermore, the scheduled breaks contributed to a noticeable reduction in eye dryness and irritation among participants. These findings emphasize the importance of incorporating behavioral strategies to address visual fatigue associated with prolonged exposure to digital screens. Moreover, promoting awareness of screen time and encouraging periodic visual breaks may significantly reduce the likelihood of developing eye strain, supporting the development of systems and interventions that aim to promote healthier digital device usage (Talens-Estarelles et al. 2023).

The effectiveness of blue-light-filtering lenses in reducing eye strain associated with prolonged screen use has been examined through a randomized controlled trial. The study involved participants who wore blue-blocking lenses while performing various dig-

ital tasks, and the results were compared with those of participants who used standard lenses. The findings indicated that the use of blue-light-filtering lenses provided modest improvements in visual comfort and helped reduce symptoms such as eye fatigue and sensitivity to glare. However, the level of effectiveness differed among participants. These results suggest that protective eyewear may help alleviate some of the negative effects of extended screen exposure. The study also contributes to the understanding of practical approaches that support eye health in digital environments and complements technological strategies designed to monitor and predict fatigue levels (Singh et al 2021).

Dandumahanti et al. (2025) developed a system for monitoring eye strain during one-hour smartphone usage through eye activity measurements. The study revealed that even short periods of continuous smartphone use can result in measurable visual fatigue. These findings highlight the importance of monitoring user behavior to detect early signs of eye strain.

Xu et al. (2026) explored methods for reducing visual fatigue by optimizing human-computer interaction (HCI) interfaces and implementing structured work-rest schedules. Using eye-tracking technology, the study found that improved interface design and scheduled breaks significantly reduce eye strain. These findings emphasize the role of ergonomic design in minimizing visual fatigue.

Environmental and Design Factors

Lu et al. (2026) examined whether urban forests can alleviate eye strain using eye-tracking metrics. The study revealed that exposure to natural environments improves eye movement stability and reduces signs of visual fatigue. Eye-tracking data showed enhanced visual comfort when participants interacted with green environments compared to urban settings. This finding suggests that environmental factors influence eye strain and

should be considered in fatigue assessment. The study provides insight into how external visual stimuli can affect ocular health, supporting the integration of contextual data in predictive systems.

The restorative effects of urban green spaces on eye strain have been investigated. The study found that natural visual environments help reduce visual fatigue and improve eye relaxation. Participants who were exposed to green spaces experienced less eye strain compared to those in artificial environments. These findings highlight the importance of environmental design in reducing digital eye strain. The results also suggest that predictive systems may be improved by including environmental and behavioral factors to enhance the accuracy of fatigue detection (Lu et al. 2025).

Li et al. (2024) developed a humidity-sensing material for monitoring eye strain through environmental and physiological conditions. The study introduced an innovative approach using sensor technology to detect changes associated with eye fatigue. Results showed that humidity levels and ocular conditions are closely related to visual discomfort. Although effective, the method requires specialized hardware, which may not be practical for everyday use. This limitation highlights the importance of developing software-based solutions using accessible data such as user behavior.

The influence of environmental factors, such as natural daylight and access to outdoor views, on eye strain and cognitive performance in office environments has been examined. Workspaces that provide natural lighting and external views were found to reduce visual fatigue and increase overall user satisfaction compared to enclosed environments that rely primarily on artificial lighting. Furthermore, individuals exposed to natural daylight demonstrated improved concentration and better task performance. These findings emphasize the significance of environmental design in promoting visual comfort and sup-

porting workplace productivity. The results also suggest that combining environmental considerations with technological interventions may strengthen strategies aimed at minimizing eye strain associated with prolonged use of digital devices (Jamrozik et al. 2019)

Tao et al. (2024) examined how different color schemes influence visual fatigue and task performance under varying conditions. The findings indicated that poor visual design and inappropriate color combinations can increase eye strain and reduce efficiency in visual tasks. The study emphasized the importance of user interface design in minimizing fatigue. This insight is relevant for developing systems that not only predict eye strain but also recommend optimized screen settings.

Vidiyanti et al. (2026) examined classroom visual fatigue and its determinants, focusing on daylighting conditions, learning media, and spatial factors within educational environments. The study revealed that insufficient natural lighting and prolonged exposure to digital learning materials significantly contribute to visual fatigue among students. Classrooms that integrate proper lighting design and optimized spatial arrangements were found to reduce eye strain and improve students' visual comfort. These findings emphasize the importance of environmental considerations in minimizing eye strain in modern digital learning settings.

Lin et al. (2026) investigated the effects of ambient illumination and mobile phone screen brightness on tear film stability, visual fatigue, and blink patterns during reading activities. The study found that excessive screen brightness combined with poor ambient lighting disrupts tear film stability and reduces blink rate, resulting in increased eye strain. The findings highlight the importance of maintaining appropriate screen brightness and lighting conditions to preserve ocular health during prolonged digital device use.

Jia et al. (2023) examined visual fatigue caused by stereoscopic displays using multimodal measurement techniques. The study found that prolonged exposure to stereoscopic disparity increases visual discomfort and contributes to eye strain. This research provides insights into how advanced display technologies may affect ocular health.

Sengsoon and Intaruk (2025) investigated the effects of light mode and dark mode display settings on visual fatigue. The study found that display modes influence visual comfort differently, with some users experiencing reduced eye strain when using dark mode. This suggests that customizable display settings can help alleviate eye strain.

Li et al. (2024) examined illumination-related fatigue using eye-tracking technology in controlled environments. The study found that poor lighting conditions significantly contribute to visual fatigue, reinforcing the importance of proper illumination in reducing eye strain.

Prevalence and Impact

Mataftsi et al. (2023) conducted a systematic review on digital eye strain in young screen users. The study summarized existing research and confirmed the high prevalence of eye strain symptoms among students and young adults. Common symptoms included dryness, headaches, blurred vision, and eye discomfort. The review emphasized the growing impact of digital device usage on ocular health. The findings strongly support the need for preventive strategies and predictive systems that can monitor and reduce eye strain in college student populations.

Moore et al. (2024) investigated the prevalence and impact of digital eye strain among working adults in the United Kingdom and Ireland. The study found that individuals who spend extended hours using computers and digital devices commonly experience symp-

toms such as eye discomfort, dryness, headaches, and blurred vision. Results also indicated that increased screen exposure was associated with reduced work productivity and decreased visual comfort throughout the day. These findings highlight the growing concern surrounding digital eye strain in modern workplaces where digital devices are essential tools for daily tasks. The study emphasizes the importance of monitoring screen exposure and implementing preventive measures to reduce visual fatigue. The insights from this research are relevant to the present study because they demonstrate how prolonged screen time can significantly affect eye health, supporting the need for systems that can predict fatigue using behavioral data such as screen usage patterns.

Research into digital eye strain indicates that susceptibility to ocular fatigue is significantly influenced by age-related physiological shifts, such as diminished tear production and slower visual accommodation. When older adults engage in frequent computer or mobile device use, these biological factors exacerbate visual discomfort compared to those with minimal screen time, proving that digital fatigue is a universal issue across all age demographics. By identifying how these individual physical characteristics interact with prolonged technology use, researchers can better develop predictive models that integrate both behavioral habits and physiological indicators (Moore et al. 2025).

Al Dandan et al. (2021) conducted a cross-sectional survey examining the prevalence of digital eye strain among radiologists who regularly analyze medical images on computer monitors. The study found that a significant percentage of participants experienced symptoms such as eye fatigue, headaches, blurred vision, and dryness due to prolonged screen exposure during diagnostic tasks. Radiologists often spend many hours focusing on digital displays, making them particularly vulnerable to visual fatigue. The findings highlight the occupational risks associated with continuous screen use and emphasize the need for effective monitoring and prevention strategies in high-screen-use profes-

sions. This study demonstrates how digital eye strain can affect performance in visually demanding tasks, reinforcing the importance of developing technologies capable of detecting early signs of fatigue.

Krupinski (2021) discussed the importance of studying eye strain among radiologists, emphasizing how prolonged exposure to digital screens may affect diagnostic accuracy and visual performance. The article explains that visual fatigue can impair concentration and reduce the ability to detect subtle details in medical images, potentially impacting the quality of clinical decision-making. The author highlighted that as healthcare increasingly relies on digital imaging technologies, understanding the effects of screen-induced fatigue becomes more critical. The study also suggests that improving work environments and developing supportive technologies may help reduce visual strain among professionals. These insights further illustrate the broader implications of digital eye strain in technology-driven fields and emphasize the value of systems capable of predicting and mitigating fatigue before it negatively affects performance.

Tan and Reyes (2026) explored the lived experiences and coping mechanisms of educators in a Philippine university dealing with digital eye strain. The qualitative study revealed that educators face a significant "digital age crisis" where constant digital demands lead to symptoms like ocular fatigue, blurred vision, and headaches, which hinder professional productivity and lesson planning. The findings highlight that digital eye strain is not merely an individual health issue but is deeply influenced by workplace conditions and institutional support. This research is particularly relevant to the present study as it provides a localized context within the Philippines, emphasizing the need for systemic interventions and ergonomic policies to manage the well-being of academic professionals in technology-driven environments.

Saldanha et al. (2021) examined how the COVID-19 pandemic influenced eye strain and dry eye symptoms due to increased reliance on digital devices for work, communication, and education. The study reported a significant rise in the prevalence of visual discomfort during the pandemic, largely attributed to extended hours spent using computers, tablets, and smartphones. Participants reported symptoms including eye fatigue, dryness, blurred vision, and headaches after prolonged digital activities. The research highlights how sudden shifts toward remote work and virtual interactions intensified the risk of digital eye strain. These findings demonstrate the importance of monitoring screen usage and implementing strategies to reduce fatigue in environments where digital technology plays a central role.

Bou Ghannam et al. (2024) investigated the effects of online learning during the COVID-19 pandemic on students' ocular health. The study found that increased screen time associated with virtual classes, assignments, and digital communication significantly contributed to symptoms of digital eye strain among students. Many participants reported experiencing headaches, blurred vision, dryness, and eye fatigue after extended periods of screen exposure. The findings highlight the impact of prolonged digital learning environments on student well-being and emphasize the need for preventive strategies to manage visual fatigue. This research strongly supports the relevance of the current study, which aims to develop a machine learning-based system capable of predicting eye strain and digital fatigue based on college students' screen time behavior.

Lindo-Cano et al. (2022) investigated the prevalence of computer vision syndrome among university students and identified a high occurrence of symptoms such as eye strain, dryness, headaches, and blurred vision. The findings showed that prolonged use of digital devices for academic purposes significantly contributes to visual discomfort. The study emphasized that students who spend extended hours on screens without proper

breaks are more likely to develop digital eye strain. This research supports the growing concern that modern educational environments increase the risk of visual fatigue, reinforcing the importance of monitoring screen time in developing predictive systems.

Muntz et al. (2022) examined the relationship between extended screen time and dry eye symptoms among young individuals. Results indicated that prolonged exposure to digital screens reduces blink rate and destabilizes the tear film, leading to dryness and discomfort. The study highlighted that younger populations are particularly vulnerable due to continuous device usage for both academic and recreational purposes. These findings provide physiological evidence linking screen time duration to eye strain, which is essential for machine learning models that aim to predict fatigue based on usage behavior.

AlQarni et al. (2023) examined the prevalence of digital eye strain among university students during virtual learning. The study found that a significant proportion of students experienced symptoms such as eye discomfort, headaches, and blurred vision due to increased screen exposure. The results highlighted that online learning environments significantly contribute to visual fatigue. This reinforces the relevance of focusing on students as the target population for predictive eye strain systems.

Victor et al. (2023) conducted a web-based cross-sectional survey assessing eye strain and perceived stress among medical students during online learning. The findings indicated a strong relationship between prolonged screen time, increased stress levels, and visual fatigue. This supports the use of survey-based methodologies in studying eye strain among students.

A review study examined how digital environments affect eye health, especially the surface of the eyes. It found that reduced blinking, poor lighting, and long periods of close

screen use are major causes of digital eye strain. The study also explained that environmental and behavioral factors can make eye discomfort worse. These findings help explain how digital habits affect eye health and support the use of behavioral data, such as screen time, in developing systems that can predict and reduce eye strain (Wolffsohn et al. 2023).

The relationship between excessive screen time and cognitive difficulties among adolescents has been examined. The findings showed that increased screen exposure is associated not only with visual fatigue but also with reduced attention and lower cognitive performance. The study suggested that prolonged use of digital devices can negatively affect both mental and visual health. These results highlight the wider impact of screen time and suggest that eye strain prediction systems may also help improve students' productivity and overall well-being (Onyeaka et al. 2022).

Chau et al. (2024) analyzed the relationship between excessive screen time and the risk of school and out-of-school injuries. The results showed that prolonged digital device usage is associated with decreased alertness and increased physical risks. Although the focus is not solely on eye strain, the study reinforces the negative consequences of excessive screen exposure. This supports the idea that monitoring screen time is crucial not only for eye health but also for general safety and performance.

Kopilaš et al. (2025) explored the relationship between visual functioning and mental health in the digital age. The study found that prolonged screen exposure negatively affects both eye health and psychological well-being, highlighting the broader impact of digital eye strain.

Pucker et al. (2024) provided updated perspectives on digital eye strain, emphasizing

its increasing prevalence due to widespread use of digital devices. The study highlighted the importance of preventive strategies and continuous monitoring to address this growing concern.

CHAPTER 3 - METHODOLOGY

This chapter presents the research design and the methods used by the researcher in conducting the study. It includes a discussion on the research methods used, research instruments, data-gathering procedures, and statistical treatment of data.

Research Method

This study uses a quantitative research method with a descriptive and predictive approach to analyze the relationship between screen time behavior and the prevalence of eye strain and digital fatigue in college students. The quantitative method is appropriate since it collects measurable data such as screen time length, session frequency, and self-reported symptoms, which can then be examined statistically and using machine learning techniques.

A cross-sectional design is utilized in this study, where data is collected from respondents at a single point in time. This approach enables the identification of patterns and relationships between screen exposure and reported eye strain symptoms without requiring long-term observation. The collected data serves as the foundation for training and evaluating the machine learning models used in the system.

Algorithm

This study utilizes machine learning classification algorithms, specifically Random Forest and XGBoost (Extreme Gradient Boosting), to predict the risk of eye strain and digital fatigue based on screen time data and user-reported symptoms. These algorithms are selected due to their high accuracy, robustness, and effectiveness in handling structured datasets.

Random Forest Algorithm

Random Forest is an ensemble learning algorithm that operates by constructing multiple decision trees during training and generating predictions based on the majority vote of these trees. Each decision tree is trained using a random subset of the dataset and a random selection of features, which enhances the diversity of the model and reduces the risk of overfitting.

In this study, the Random Forest algorithm is used to analyze input data such as screen time duration, frequency of device use, and reported eye strain symptoms. The algorithm identifies patterns in the data and classifies the level of eye strain into categories such as low, moderate, or high.

The use of Random Forest is advantageous due to its ability to handle nonlinear relationships, manage missing or inconsistent data, and provide stable and reliable predictions. Furthermore, its ensemble nature improves generalization performance, making it suitable for real-world applications involving user generated data.

Extreme Gradient Boosting (XGBoost) Algorithm

XGBoost, or Extreme Gradient Boosting, is an advanced machine learning algorithm based on gradient boosting. It builds models step by step, where each new model corrects the errors of the previous one. This process improves the accuracy of predictions.

In this study, XGBoost is used to better understand the relationship between screen time behavior and symptoms of digital eye strain. It also includes techniques to prevent overfitting, which helps the model perform well on new data.

XGBoost is selected for its computational efficiency, scalability, and high performance in classification tasks. It is particularly effective in handling structured datasets and is widely recognized for achieving superior results in machine learning competitions and real-world applications.

System Design

This study follows the Agile Software Development Methodology in developing the web-based eye strain and digital fatigue prediction system. Agile is selected because it allows continuous improvement, flexibility, and iterative development, which are essential in building a system that involves user feedback and machine learning model refinement.

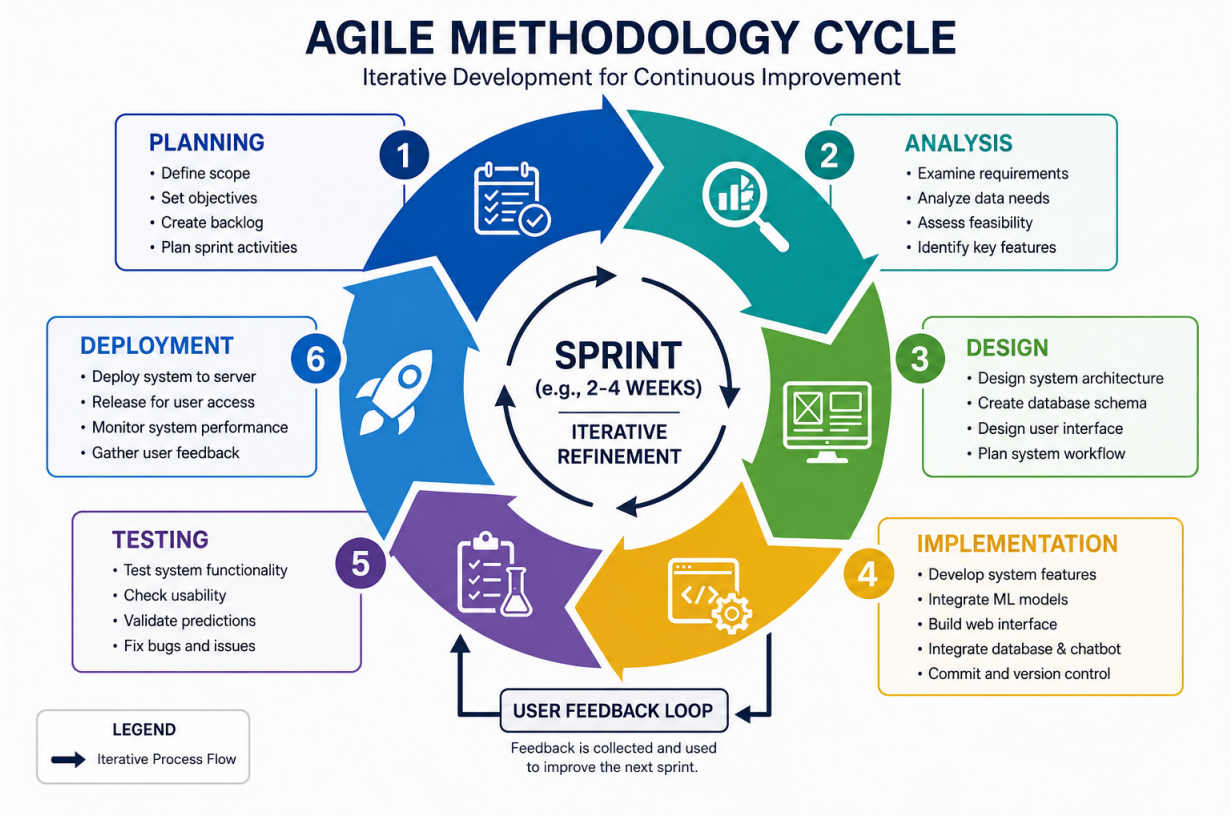


Figure 3.1: Agile Development

Planning

During the Planning Phase, the researchers identified the main problem addressed by the study and defined the objectives and scope of the proposed system. The required system inputs, such as screen time duration and symptom-related data, as well as the expected prediction output, were also determined. In addition, the necessary tools, the structure of the data set, and the overall workflow of the system were planned during this phase.

The researchers also developed the survey questionnaire for data collection and consulted eye care experts and potential respondents to gather relevant insights and recommendations. These inputs served as important references to guide the development of the proposed system.

Analysis

The Analysis Phase focused on examining the data requirements and evaluating the feasibility of the proposed system. Relevant inputs necessary for prediction, such as screen time duration, frequency of device usage, and self-reported symptoms associated with eye strain and digital fatigue, were identified and analyzed. Functional and non-functional requirements were likewise assessed to ensure the effectiveness, reliability, usability, and overall performance of the system. The requirements for the AI chat-bot, admin panel, and database management were also analyzed during this phase.

Design

In the Design Phase, the overall structure and architecture of the proposed system were developed. This phase involved the design of the system architecture, database schema, and user interface. Interface layouts and wire frames were also created to ensure that the system is user-friendly, accessible, and easy to navigate.

In addition, the researchers designed the system workflow and functionality to ensure the efficient processing of user inputs, machine learning-based prediction, and generation of results. The interface and functionality of the AI chat bot and admin panel were likewise designed during this phase. This stage established the foundational design that guided the development and implementation of the system

Implementation

The Implementation Phase involved the actual development and integration of the proposed system. During this stage, the web-based interface was developed, and the machine learning algorithms, specifically Random Forest and XGBoost, were integrated into the system. Data preprocessing, model training, and prediction functionalities were also implemented to ensure proper system operation. The researchers also included a machine learning retraining capability wherein newly uploaded user data stored in the Supabase database may be fetched and utilized for future retraining of the machine learning model. This process allows the system to adapt to additional datasets and potentially improve prediction accuracy and overall system performance over time. In addition, the AI chat-bot feature, admin panel, and Supabase backend integration were developed to support user assistance, record management, and secure data storage. The system also supports light mode and dark mode features to enhance user experience, interface accessibility, and visual comfort based on user preference and viewing conditions.

Testing

During the Testing Phase, the developed system was evaluated in terms of functionality, usability, and prediction accuracy. Errors, inconsistencies, and performance issues were identified and addressed through continuous testing and refinement. User feedback was also collected to further improve the system's overall performance and effectiveness.

The AI chat-bot responses, admin panel functionalities, and database operations were likewise tested to ensure proper operation and usability.

Deployment

Finally, the Deployment Phase involved making the completed system accessible through a web browser for actual use and evaluation. Users were able to input screen time data and receive real-time predictions of their eye strain and digital fatigue risk levels. This phase ensured that the system was fully operational and ready for implementation.

Theoretical Framework

This study is anchored on the principles of Machine Learning, Human-Computer Interaction (HCI), and Digital Eye Strain Theory, which serve as the foundation for the development of the proposed system.

Machine Learning provides the main computational basis of the study. It enables the system to analyze patterns in user-generated data and predict eye strain and digital fatigue. Using classification algorithms such as Random Forest and XGBoost, the system processes structured data and produces predictions based on the relationship between screen time behavior and reported symptoms.

Human-Computer Interaction (HCI) focuses on the usability and design of the system. It emphasizes the creation of a simple and accessible interface that allows college students to input data easily and understand the results clearly. The application of HCI principles ensures that the system is user-friendly and effective.

In addition, the study is guided by the Digital Eye Strain Theory, which explains the

discomfort caused by prolonged exposure to digital screens. It identifies factors such as reduced blinking, continuous focus on bright screens, improper screen brightness, and long periods of device use. These factors are linked to symptoms such as eye dryness, headaches, blurred vision, and eye fatigue. In this study, the theory serves as the conceptual basis for understanding how screen time behavior affects the risk of digital eye strain among college students.

Conceptual Framework

The conceptual framework of the study illustrates the overall process and interaction of components within the proposed Machine Learning-Based Eye Strain and Digital Fatigue Prediction System. The framework begins with the users, particularly college students, who provide relevant input data such as daily screen time duration, frequency of device usage, and self-reported symptoms associated with eye strain and digital fatigue.

The collected data is processed through the web-based system, where data collection, preprocessing, feature encoding, and prediction procedures are performed. During this stage, the machine learning algorithms, specifically Random Forest and XGBoost, analyze the processed data to identify patterns and predict the level of risk associated with eye strain and digital fatigue.

The system also integrates an AI chat-bot feature that provides general information, guidance, and recommendations related to healthy screen usage and eye strain management. In addition, an admin panel is incorporated to manage user records, monitor prediction results, and oversee system operations. The collected information and generated predictions are stored using Supabase for database management and secure data storage. Based on the generated predictions, the system provides outputs such as the predicted risk level, eye strain assessment, and recommended actions to help reduce

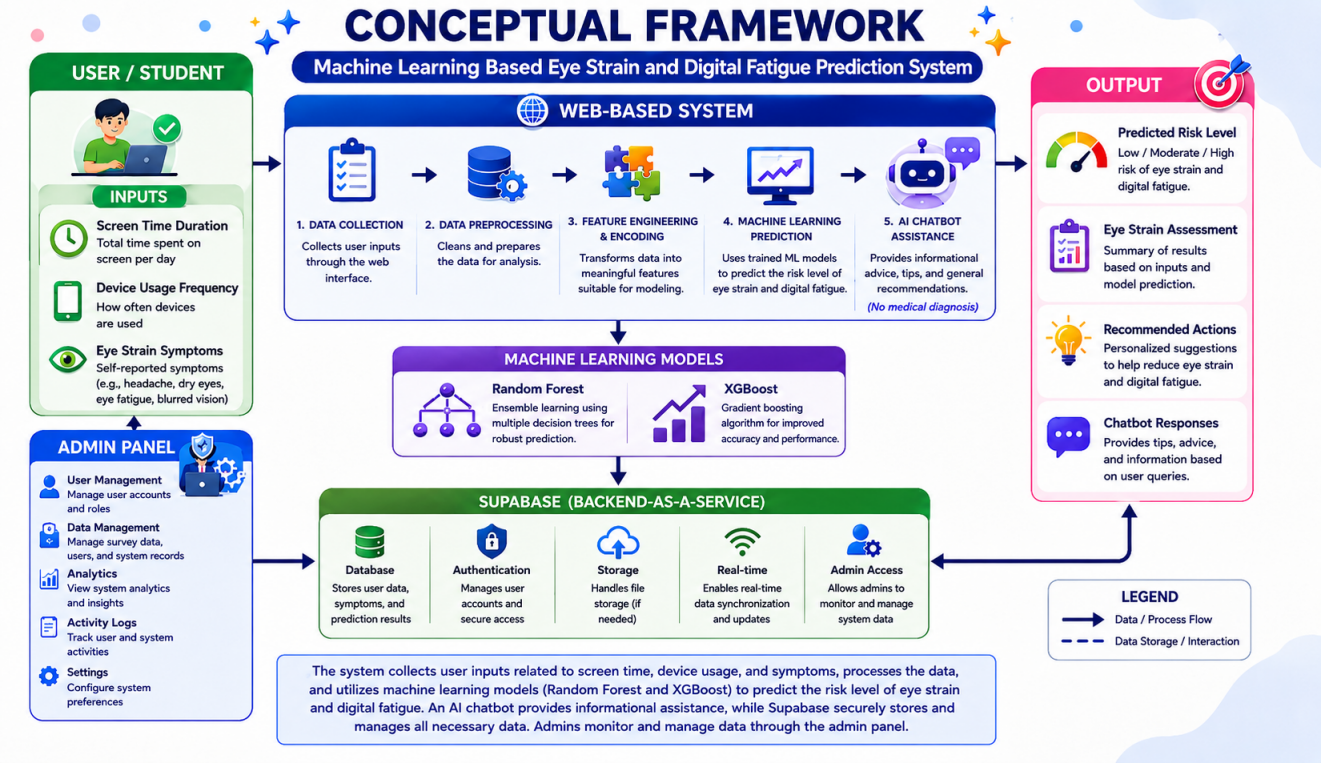


Figure 3.2: Conceptual Framework of the Proposed Machine Learning-Based Eye Strain and Digital Fatigue Prediction System

screen-related discomfort. The framework demonstrates how user inputs are transformed into meaningful predictive insights through the integration of machine learning techniques and web-based technologies.

Statistical Treatment of Data

To evaluate the performance of the machine learning models used in this study, several statistical metrics are applied, including Accuracy, Precision, Recall, and F1-Score. These metrics are derived from the confusion matrix, which consists of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Accuracy measures the overall correctness of the model's predictions and is computed using the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.1}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Precision is used to determine the proportion of correctly predicted positive cases and is expressed as:

$$Precision = \frac{TP}{TP + FP} \tag{3.2}$$

Recall measures the ability of the model to correctly identify actual positive cases and is calculated using the formula:

$$Recall = \frac{TP}{TP + FN} \tag{3.3}$$

Meanwhile, the F1-Score provides a balanced evaluation between Precision and Recall and is computed as:

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3.4}$$

For system evaluation, the study utilizes a Likert scale to assess software quality attributes such as functionality and usability. The weighted mean is used to analyze respondents' evaluations and is represented by the formula:

$$\bar{x} = \frac{\sum fx}{N} \tag{3.5}$$

Where:

- $\bar{x}$ = Weighted Mean
- f = Frequency of responses
- x = Scale value
- N = Total number of responses

Additionally, Analysis of Variance (ANOVA) may be applied to determine whether significant differences exist among respondents' evaluations of the system.

Population and Sampling

The target population of this study consists of college students who frequently use digital devices for both academic and personal purposes. A non-probability sampling technique, specifically convenience sampling, is employed in selecting respondents. Participants are chosen based on their availability and willingness to provide the necessary data for the study. The research aims to collect responses from a sufficient number of participants to ensure reliable model training and evaluation. To qualify for participation, respondents must be currently enrolled college students, regularly use digital devices, and be willing to provide information regarding their screen time behavior and self-reported symptoms related to eye strain and digital fatigue.

Data Gathering

The data collection process begins with the preparation of a structured questionnaire designed to gather information on respondents' screen time behavior and symptoms of eye strain and digital fatigue. The questionnaire includes items related to daily screen time duration, frequency of device use, and commonly experienced symptoms such as eye dryness, headaches, and blurred vision. After development, the survey is distributed to college students through online platforms such as Google Forms and the web-based system developed for the study.

Respondents then provide their answers by reporting their screen usage patterns and indicating the symptoms they experience. The collected data is reviewed and validated to ensure quality, and incomplete or inconsistent responses are removed. After validation, the reliable data is securely stored in the system database, making it ready for further processing and machine learning model training.

Research Ethics

The study adheres to established ethical standards throughout data collection, system development, and evaluation. Participation is entirely voluntary, and respondents may join or withdraw at any time without penalty. Prior to data collection, participants receive information regarding the study's purpose, procedures, and the intended use of collected data.

To ensure privacy, all personal information provided by respondents is handled with strict confidentiality. Data is collected anonymously whenever feasible, and no identifying details are included in any report, publication, or presentation associated with the study. All collected information is used exclusively for academic and research purposes.

The study only collects data relevant to screen time behavior and self-reported symptoms associated with eye strain and digital fatigue. No highly sensitive personal, financial, or medical records are requested from participants. Appropriate security measures are also applied to stored data to minimize unauthorized access. Through these practices, the study ensures minimal risk to respondents while maintaining the integrity and credibility of the research process.

CHAPTER 4 - RESULTS AND DISCUSSIONS

In this chapter, the results of the findings, analysis, and interpretation of data are presented. The data were collected and then processed in response to the problems posed in the statement of the problem.

Findings/Answers for each statement of the problem

1. How do the researchers plan to create a solution on water scarcity?

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2. How do the researchers develop to the problem on greywater discharge harvesting?

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3. How do the researchers evaluate the proposed project Greywater Recycling System by using ISO 25010 in terms of the following aspect?

Functional Suitability;

Table 4.1: Evaluation of the System's Functional Suitability by Researcher

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Performance Efficiency;

Table 4.2: Evaluation of the System's Performance Efficiency by Researcher

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Reliability?

Table 4.3: Evaluation of the System's Reliability by Researcher

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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4. How do the households evaluate the proposed project Greywater Recycling System by using ISO 25010 in terms of the following aspect?

Usability;

Table 4.4: Evaluation of the System's Usability by Household

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Reliability?

Table 4.5: Evaluation of the System's Reliability by Household

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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5. How do the IT Professionals evaluate the proposed project Greywater Recycling System by using ISO 25010 in terms of the following aspect?

Functional Suitability;

Table 4.6: Evaluation of the system's Functional Suitability by IT Professionals

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Performance Efficiency;

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Table 4.7: Evaluation of the System's Performance Efficiency by IT Professionals

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Usability;

Table 4.8: Evaluation of the System's Usability by IT Professionals

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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Reliability?

Table 4.9: Evaluation of the System's Reliability by IT Professionals

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
The proposed system sends and receives data anytime and anywhere given that the technology used in the system is IoT.	1	1	1	1	1	1	100%
Performance Efficiency							100%

*T = Trial, true = 1, false = 0

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CHAPTER 5 - SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

In this chapter, the researcher summarizes, concludes, and recommends the outcome of detecting and analyzing vitreoretinal conditions using convolutional neural network types based on the accomplished experiments and testing.

Summary of Basic Findings

1. Summary of findings in SOP 1

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2. Summary of findings in SOP 2

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3. Summary of findings in SOP 3

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4. Summary of findings in SOP 4

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5. Summary of findings in SOP 5

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Conclusions

Based on the results of the study, the researchers conclude the following:

1. Conclusions in SOP 1

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2. Conclusions in SOP 2

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3. Conclusions in SOP 3

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4. Conclusions in SOP 4

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5. Conclusions in SOP 5

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Recommendations

Based on the results of the study, the researchers recommend the following:

1. Recommendations in SOP 1

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2. Recommendations in SOP 2

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3. Recommendations in SOP 3

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4. Recommendations in SOP 4

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5. Recommendations in SOP 5

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